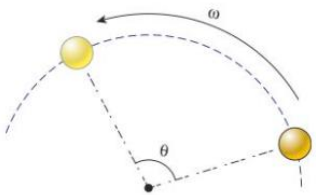
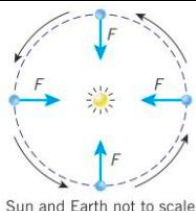


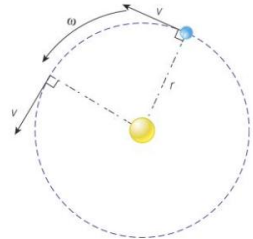
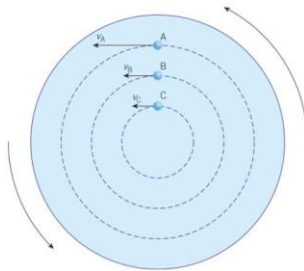
1. Lesson 1 Angular velocity and radian

Radian	A radian is the angle subtended by a circular arc with a length equal to the radius of the circle. This is an angle of approximately 57.3° for any circle. $\text{angle in radians} = \frac{\text{arc length}}{\text{radius}}$
Angle in radians	$\text{angle in radians} = \frac{2\pi r}{r} = 2\pi \text{ radians}$ Therefore 360° is equal to 2π radians, or about 6.3 radians.
Conversion degrees into radians	Divide the angle in degrees by $\frac{180}{\pi}$.
Angular velocity	The rate of change of angle for an object moving in a circular path – symbol ω $\omega = \frac{\theta}{t}$ or $\omega = \frac{2\pi}{T} = 2\pi f$ T – one period, second s ω – angular velocity is measured in radians per second f – frequency is the reciprocal of the period, measured in hertz (Hz)
	 The faster the object moves, the greater the angle moved through (subtended) in a given time.

Angular velocity units	Degrees per second ($^\circ\text{s}^{-1}$) Revolutions per second (revs^{-1}) Revolutions per minute (rpm) Radians per second (rads^{-1}) – A level physics course
-------------------------------	--

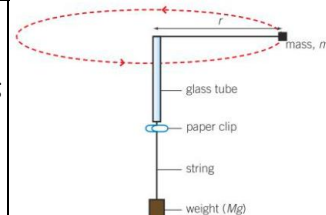
2. Lesson 2 Angular acceleration

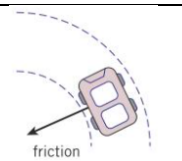
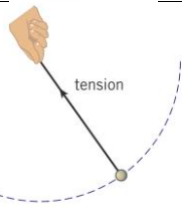
Centripetal force	A force that keeps a body moving with a constant speed in a circular path. It is always perpendicular to the velocity of the object. This means that this force has no component in the direction of motion and so no work is done on the object. As a result its speed remains constant.
Centripetal (ancient greek)	means 'centre-seeking'.
Centripetal force example	 Sun and Earth not to scale - gravitational attraction for a satellite in orbit around the planet - friction for a car going around a bend - tension in a string when a yo-yo is swung around in a vertical circle
Circular motion net force	In all cases of circular motion, there must be a net force towards the centre of the circle.

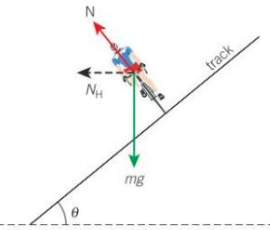
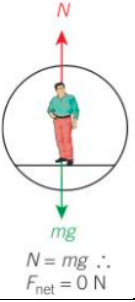
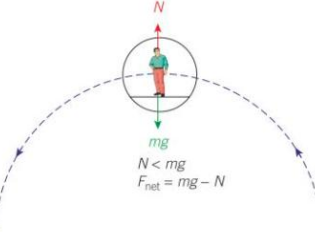
Linear velocity	At any point on a circular path, the linear velocity is always at a tangent to the circular path (at 90° to the radius). 
Speed – circular motion	$\text{speed} = \frac{\text{distance travelled}}{\text{time taken}}$ $v = \frac{2\pi r}{T}$
Angular velocity	$\omega = \frac{2\pi}{T}$ $v = r\omega$
Linear velocity VS angular velocity	For objects with the same angular velocity, the linear velocity at any instant is directly proportional to the radius. Example 
Centripetal acceleration	The acceleration of any object travelling in a circular path at constant speed, which always acts towards the centre of the circle.

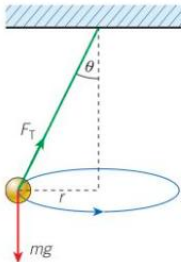
Centripetal acceleration Direction	Centripetal acceleration always acts towards the centre of the circular path.
Centripetal acceleration Equation	$a = \frac{v^2}{r}$ Combining this equation with $v = r\omega$ Gives an alternative expression $a = \omega^2 r$

3. Lesson 3 Exploring centripetal forces

Centripetal force	A force that keeps a body moving with a constant speed in a circular path
Centripetal force Equation	Combining $F = ma \quad \text{and} \quad a = \frac{v^2}{r}$ we can determine an expression for centripetal force $F = \frac{mv^2}{r}$ m – mass, in kg r – radius, in m $F = m\omega^2 r$
Investigating circular motion	

Whirling bung	As the bung swung in a horizontal circle the suspended weight remains stationary as long as the force it provides is equal to the centripetal force required to make the bung travel in the circular path. The paperclip acts as a marker to make this movement clearer. The weight and thus the centripetal force required for different masses, radii, and speeds (calculated from angular velocity and radius) can then be investigated.
Centripetal force Friction	
Centripetal force Tension	
Centripetal force Gravitational force	
Banked surfaces	The tracks in modern velodromes are banked up to angles of 45° so that track cyclists can travel at higher speeds. On the banked part of the track a horizontal component of the normal contact force, together with the frictional force from the tyres, provides the centripetal force required to follow the circular path at such high speed.

	 ▲ Figure 5 A horizontal component of the normal contact force N acts towards the centre of the circle. $N_H = N \sin \theta$.
	The friction between the tyres and track would not be sufficient to allow the cyclist to travel at that speed on a flat surface.
Ferris wheel	When the capsule is stationary the normal contact force N is equal to your weight.  $N = mg \therefore F_{\text{net}} = 0 \text{ N}$
	When the wheel rotates a net force is required in order for you to travel in a circular path. At the top of the ride N reduces, resulting in a net force towards the centre of the circle. This also gives you a feeling of slightly reduced weight.  $N < mg$ $F_{\text{net}} = mg - N$

Conical pendulum	A conical pendulum rotates at constant speed, describing a horizontal circle. The time taken to complete each rotation depends only on the length of the pendulum string and the gravitational field strength.  From this equation we can see that the angle of the pendulum is not affected by the mass of the pendulum bob. $\tan \theta = \frac{v^2}{rg}$
Conical pendulum uses	To calibrate the tracking of sensitive telescopes that move slowly against the Earth's rotation to ensure they remain fixed on a certain point in the sky.